



expEYES

A Low-cost Platform for PC-interfaced Science Experiments

There is nothing new about computer-interfaced science experiments, but how often do you come across an experiment platform that costs as little as Rs 3000? Inter University Accelerator Centre (IUAC) has developed one such framework that can also be used as an electronics developers' kit

Schools and colleges often focus on theory, forgetting that practical experiments teach students much more. Members of the Inter University Accelerator Centre (IUAC), an autonomous research institute that provides research facilities to universities, realised the need for experiments in education when teachers and students from other colleges visited their centre and tried to use a particle accelerator. They were not familiar with computer-controlled experiments that use machines to collect data.

Dr Ajith Kumar, a scientist at IUAC, wished to provide teachers and students with exposure to this technology and started designing a low-cost device that measured and collected data, and displayed the results graphically on the computer screen.

Along with members of the electronics department, he developed Phoenix expEYES—a combination of hardware and a software framework for computer-interfaced science experiments that doesn't require the user to get into the details of electronics or computer programming.

"We decided to create something on a smaller scale and

introduce the idea of a low-cost computer-interfaced platform for science experiments," comments Dr Kumar.

A platform for experiments

The equipment places emphasis on leveraging the power of personal computers for experiment control, data acquisition and processing. Phoenix expEYES enables one to carry out a variety of experiments using the same inexpensive hardware with different sensor elements and software. The design also enables the platform to be used as an electronics developers' kit, as it can be used to develop and learn microcontroller programming. Engineering students can play around with the platform or use it as a case study to design similar applications. Since the code and design are open, one can get any details required from IUAC's website.

What sets it apart

Dr Kumar emphasises, "Our main aim was to make the equipment affordable to educational institutes. Phoenix ExpEYES is priced at around Rs 3000, and a similar product